

Project Resources

CIS 5820 Full-Stack Development Project

A curated set of free, primarily official resources to help you learn what you need for your project. These are starting points, not a required reading list. Use what is relevant to what you are building, and treat learning from any of these as legitimate project progress.

Please note: I used a variety of tools to find these, and I have tried to verify each, but I have not personally used everyone of them. If you find an invalid resource or broken link, please let me know.

Learning Android and Kotlin

- **[Android Kotlin Fundamentals](https://developer.android.com/courses/kotlin-fundamentals/course)** — Google's official course: build five Android apps in Kotlin while learning architecture components, Room, RecyclerView, and loading data from the internet. (<https://developer.android.com/courses/kotlin-fundamentals/course>)
- **[Learn Kotlin for Android](https://developer.android.com/kotlin/campaign/learn)** — Google's official landing page collecting Kotlin-for-Android training at every level, from little to no programming experience up through coroutines. (<https://developer.android.com/kotlin/campaign/learn>)
- **[Android Developers: Courses](https://developer.android.com/courses)** — The official index of all of Google's free Android training courses, organized by topic and skill level. (<https://developer.android.com/courses>)
- **[Android Codelabs](https://developer.android.com/codelabs)** — Official, guided, hands-on tutorials from Google. Short and focused; a good way to learn one specific piece (a library, a pattern, a component) quickly. (<https://developer.android.com/codelabs>)
- **[Kotlin Documentation](https://kotlinlang.org/docs/home.html)** — The official Kotlin language documentation and tour, maintained by JetBrains. (<https://kotlinlang.org/docs/home.html>)
- **[Master Kotlin & Android \(60-Hour Course\)](https://www.freecodecamp.org/news/master-kotlin-and-android-60-hour-course/)** — A free, long-form video course on freeCodeCamp's YouTube channel covering Kotlin, Android Studio, Room, Firebase, MVVM, and more from the ground up. (<https://www.freecodecamp.org/news/master-kotlin-and-android-60-hour-course/>)

Mobile UI/UX Design

- **[Material Design 3](https://m3.material.io/)** — Google's official design system: components, layout guidance, color, typography, and interaction patterns for Android apps. (<https://m3.material.io/>)
- **[Material Design for Android](https://developer.android.com/develop/ui/views/theming/look-and-feel)** — The official developer documentation on applying Material theming to your app, including light and dark themes. (<https://developer.android.com/develop/ui/views/theming/look-and-feel>)
- **[Android Design Guidelines \(CodePath\)](https://guides.codepath.org/android/Android-Design-Guidelines)** — A free, practical overview of Android design principles with links out to the official Material references, aimed at students. (<https://guides.codepath.org/android/Android-Design-Guidelines>)

Backend and APIs for Mobile Apps

- **[Retrofit](https://square.github.io/retrofit/)** — The standard, widely used type-safe HTTP client for Android. Official documentation and setup guide. (<https://square.github.io/retrofit/>)
- **[Networking with Android \(Official Guide\)](#)** — Google's own introduction to making network

requests from an Android app, including background work and connectivity checks.

(<https://developer.android.com/training/basics/network-ops>)

- **[Firebase](#)** — Google's backend-as-a-service platform: hosted databases, authentication, storage, and messaging, useful if you do not want to run your own server. (<https://firebase.google.com/>)
- **[Public APIs](#)** — A large, community-maintained list of free public APIs, useful if your app needs to pull in real data (weather, sports, recipes, and many other categories). (<https://github.com/public-apis/public-apis>)

Databases and Local Storage

- **[Room Persistence Library](#)** — The official guide to Room, Android's recommended library for local databases, built on top of SQLite. (<https://developer.android.com/training/data-storage/room>)
- **[SQLite](#)** — Documentation for SQLite itself, the database engine Room is built on. Useful if you need to understand what is happening underneath Room. (<https://sqlite.org/docs.html>)
- **[Preferences DataStore](#)** — Google's official guide to DataStore, the modern, simpler alternative to SharedPreferences for storing small amounts of data, such as settings. (<https://developer.android.com/topic/libraries/architecture/datastore>)

Testing and Debugging Mobile Apps

- **[Test Your App on Android \(Official Overview\)](#)** — Google's starting point for testing on Android: what to test, and how unit tests, integration tests, and UI tests differ. (<https://developer.android.com/training/testing>)
- **[AndroidX Test](#)** — Official documentation for the AndroidX Test libraries, the foundation most Android testing tools are built on. (<https://developer.android.com/jetpack/androidx/test>)
- **[Espresso](#)** — Google's official guide to Espresso, the standard framework for writing UI tests that interact with your app's screens. (<https://developer.android.com/training/testing/espresso>)

Publishing and Distribution

- **[Google Play Console Help](#)** — Google's official support hub for publishing an app: account setup, store listings, policies, and review. (<https://support.google.com/googleplay/android-developer>)
- **[Launch Checklist \(Official\)](#)** — Google's own pre-launch checklist covering technical quality, store listing, and policy compliance before you publish. (<https://developer.android.com/distribute/best-practices/launch/launch-checklist>)

General Software Engineering and Project Management

- **[Pro Git](#)** — The complete, free, official Git book. Everything from first commit to branching workflows, available to read online. (<https://git-scm.com/book/en/v2>)
- **[Git Basics \(git-scm.com\)](#)** — The official Git project's own short introduction and command reference, a faster read than the full book. (<https://git-scm.com/learn>)
- **[Atlassian Git Tutorials](#)** — A free, well-organized set of tutorials covering Git concepts and common workflows, with clear diagrams. (<https://www.atlassian.com/git/tutorials>)
- **[The Scrum Guide](#)** — The short, official definition of Scrum, a common way development teams plan and structure work in short cycles. Useful even outside a team setting, for structuring your own milestones. (<https://scrumguides.org/>)

- **Technical Writing Courses (Google)** — Google's own free courses on writing clearly about technical work, directly useful for your documentation form and README.
(<https://developers.google.com/tech-writing>)

If you find a resource that belongs on this list, tell me. This list will grow across semesters.